

Roots (Part II)

Review: Multiplying Binomials

Example I)

$$(x+2)(x-5)$$

$$x^2 - 5x + 2x - 10$$

$$\boxed{x^2 - 3x - 10}$$

Example III)

$$(x+9)(x+8)$$

$$x^2 + 8x + 9x + 72$$

$$\boxed{x^2 + 17x + 72}$$

Example V)

$$(\sqrt{3}-2)(\sqrt{3}+7)$$

$$\sqrt{3} \cdot \sqrt{3} + 7\sqrt{3} - 2\sqrt{3} - 2(7)$$

$$\sqrt{3} \cdot 3 + 7\sqrt{3} - 2\sqrt{3} - 14$$

$$\sqrt{9} + 7\sqrt{3} - 2\sqrt{3} - 14$$

$$3 + 5\sqrt{3} - 14$$

$$\boxed{5\sqrt{3} - 11}$$

FOIL

Example II)

$$(3x-2)(4x-5)$$

$$12x^2 - 15x - 8x + 10$$

$$\boxed{12x^2 - 23x + 10}$$

Example IV)

$$(6x-7)(5x+3)$$

$$30x^2 + 18x - 35x - 21$$

$$\boxed{30x^2 - 17x - 21}$$

Example VI)

$$(\sqrt{6}+2)(\sqrt{6}+4)$$

$$\sqrt{6} \cdot \sqrt{6} + 4\sqrt{6} + 2\sqrt{6} + 2(4)$$

$$\sqrt{6} \cdot 6 + 6\sqrt{6} + 8$$

$$\sqrt{36} + 6\sqrt{6} + 8$$

$$6 + 6\sqrt{6} + 8$$

$$\boxed{14 + 6\sqrt{6}}$$

Rationalizing Two-Term Denominators

① $\frac{3}{4+\sqrt{5}}$

$$\frac{3}{(4+\sqrt{5})(4-\sqrt{5})}$$

$$\frac{3(4-\sqrt{5})}{(4+\sqrt{5})(4-\sqrt{5})}$$

$$\frac{12-3\sqrt{5}}{16-4\sqrt{5}+4\sqrt{5}-\sqrt{5}\sqrt{5}}$$

$$\frac{12-3\sqrt{5}}{16-\sqrt{5} \cdot 5}$$

$$\frac{12-3\sqrt{5}}{16-\sqrt{25}}$$

$$\frac{12-3\sqrt{5}}{16-5}$$

$$\boxed{\frac{12-3\sqrt{5}}{11}}$$

② $\frac{1}{\sqrt{3}-\sqrt{7}}$

$$\frac{1}{(\sqrt{3}-\sqrt{7})(\sqrt{3}+\sqrt{7})}$$

$$\sqrt{a}\sqrt{b} = \sqrt{ab}$$

$$\frac{\sqrt{3}+\sqrt{7}}{\sqrt{3}\sqrt{3}+\sqrt{3}\sqrt{7}-\sqrt{7}\sqrt{3}-\sqrt{7}\sqrt{7}}$$

$$\frac{\sqrt{3}+\sqrt{7}}{\sqrt{3} \cdot 3 + \sqrt{3} \cdot 7 - \sqrt{7} \cdot 3 - \sqrt{7} \cdot 7}$$

$$\frac{\sqrt{3}+\sqrt{7}}{\sqrt{9} + \sqrt{21} - \sqrt{21} - \sqrt{49}}$$

$$\frac{\sqrt{3}+\sqrt{7}}{3-7}$$

$$\boxed{\frac{\sqrt{3}+\sqrt{7}}{-4}}$$

$$\textcircled{3} \frac{8}{9-\sqrt{2}}$$

$$\frac{8(9+\sqrt{2})}{(9-\sqrt{2})(9+\sqrt{2})}$$

$$\frac{8(9+\sqrt{2})}{(9-\sqrt{2})(9+\sqrt{2})}$$

$$\frac{72+8\sqrt{2}}{81+9\sqrt{2}-9\sqrt{2}-\sqrt{2}\sqrt{2}}$$

$$\frac{72+8\sqrt{2}}{81-\sqrt{2}\cdot 2}$$

$$\frac{72+8\sqrt{2}}{81-\sqrt{4}}$$

$$\frac{72+8\sqrt{2}}{81-2}$$

$$\boxed{\frac{72+8\sqrt{2}}{79}}$$

$$\textcircled{4} \frac{10}{\sqrt{7}+\sqrt{5}}$$

$$\frac{10(\sqrt{7}-\sqrt{5})}{(\sqrt{7}+\sqrt{5})(\sqrt{7}-\sqrt{5})}$$

$$\frac{10\sqrt{7}-10\sqrt{5}}{\sqrt{7}\sqrt{7}-\sqrt{7}\sqrt{5}+\sqrt{5}\sqrt{7}-\sqrt{5}\sqrt{5}}$$

$$\frac{10\sqrt{7}-10\sqrt{5}}{\sqrt{7}\cdot 7-\sqrt{7}\cdot 5+\sqrt{7}\cdot 5-\sqrt{5}\cdot 5}$$

$$\frac{10\sqrt{7}-10\sqrt{5}}{\sqrt{49}-\sqrt{35}+\sqrt{35}-\sqrt{25}}$$

$$\frac{10\sqrt{7}-10\sqrt{5}}{7-5}$$

$$\frac{10\sqrt{7}-10\sqrt{5}}{2}$$

$$\boxed{5\sqrt{7}-5\sqrt{5}}$$

$$\textcircled{5} \frac{\sqrt{6}-5}{\sqrt{2}+3}$$

$$\frac{(\sqrt{6}-5)(\sqrt{2}-3)}{(\sqrt{2}+3)(\sqrt{2}-3)}$$

$$\frac{(\sqrt{6}-5)(\sqrt{2}-3)}{(\sqrt{2}+3)(\sqrt{2}-3)}$$

$$\frac{\sqrt{12}-3\sqrt{6}-5\sqrt{2}+15}{\sqrt{4}-3\sqrt{2}+3\sqrt{2}-9}$$

$$\frac{\sqrt{4}\cdot 3-3\sqrt{6}-5\sqrt{2}+15}{2-9}$$

$$\frac{\sqrt{4}\cdot \sqrt{3}-3\sqrt{6}-5\sqrt{2}+15}{-7}$$

$$\frac{2\sqrt{3}-3\sqrt{6}-5\sqrt{2}+15}{-7}$$

$$\boxed{\frac{3\sqrt{6}+5\sqrt{2}-2\sqrt{3}-15}{7}}$$

$$\textcircled{6} \frac{\sqrt{7}+3}{5-\sqrt{10}}$$

$$\frac{(\sqrt{7}+3)(5+\sqrt{10})}{(5-\sqrt{10})(5+\sqrt{10})}$$

$$\frac{5\sqrt{7}+\sqrt{70}+15+3\sqrt{10}}{25+5\sqrt{10}-5\sqrt{10}-\sqrt{100}}$$

$$\frac{5\sqrt{7}+\sqrt{70}+15+3\sqrt{10}}{25-10}$$

$$\boxed{\frac{5\sqrt{7}+\sqrt{70}+15+3\sqrt{10}}{15}}$$

$$\textcircled{7} \frac{\sqrt{2} + \sqrt{6}}{2\sqrt{3} - \sqrt{7}}$$

$$\frac{(\sqrt{2} + \sqrt{6})(2\sqrt{3} + \sqrt{7})}{(2\sqrt{3} - \sqrt{7})(2\sqrt{3} + \sqrt{7})}$$

$$\frac{2\sqrt{6} + \sqrt{14} + 2\sqrt{18} + \sqrt{42}}{4\sqrt{9} + 2\sqrt{21} - 2\sqrt{21} - \sqrt{49}}$$

$$\frac{2\sqrt{6} + \sqrt{14} + 2\sqrt{9 \cdot 2} + \sqrt{42}}{4(3) - 7}$$

$$\frac{2\sqrt{6} + \sqrt{14} + 2\sqrt{9} \cdot \sqrt{2} + \sqrt{42}}{12 - 7}$$

$$\frac{2\sqrt{6} + \sqrt{14} + 2 \cdot 3 \cdot \sqrt{2} + \sqrt{42}}{5}$$

$$\boxed{\frac{2\sqrt{6} + \sqrt{14} + 6\sqrt{2} + \sqrt{42}}{5}}$$

$$\textcircled{8} \frac{\sqrt{10} - 3\sqrt{5}}{\sqrt{2} - \sqrt{3}}$$

$$\frac{(\sqrt{10} - 3\sqrt{5})(\sqrt{2} + \sqrt{3})}{(\sqrt{2} - \sqrt{3})(\sqrt{2} + \sqrt{3})}$$

$$\frac{\sqrt{20} + \sqrt{30} - 3\sqrt{10} - 3\sqrt{15}}{\sqrt{4} + \sqrt{6} - \sqrt{6} - \sqrt{9}}$$

$$\frac{\sqrt{4 \cdot 5} + \sqrt{30} - 3\sqrt{10} - 3\sqrt{15}}{2 - 3}$$

$$\frac{\sqrt{4} \cdot \sqrt{5} + \sqrt{30} - 3\sqrt{10} - 3\sqrt{15}}{-1}$$

$$\boxed{-2\sqrt{5} - \sqrt{30} + 3\sqrt{10} + 3\sqrt{15}}$$

Rationalizing Algebraic Denominators

$$\textcircled{9} \frac{3}{\sqrt{5x}}$$

$$\frac{3 \cdot \sqrt{5x}}{\sqrt{5x} \cdot \sqrt{5x}}$$

$$\frac{3\sqrt{5x}}{\sqrt{5x \cdot 5x}}$$

$$\frac{3\sqrt{5x}}{\sqrt{25x^2}}$$

$$\frac{3\sqrt{5x}}{\sqrt{25} \cdot \sqrt{x^2}}$$

$$\boxed{\frac{3\sqrt{5x}}{5x}}$$

$$\textcircled{10} \frac{2x}{7\sqrt{3x}}$$

$$\frac{2x \cdot \sqrt{3x}}{7\sqrt{3x} \cdot \sqrt{3x}}$$

$$\frac{2x\sqrt{3x}}{7(3x)}$$

$$\boxed{\frac{2x\sqrt{3x}}{21x}}$$

$$\textcircled{11} \frac{8}{\sqrt{x}}$$

$$\frac{8 \cdot \sqrt{x}}{\sqrt{x} \cdot \sqrt{x}}$$

$$\boxed{\frac{8\sqrt{x}}{x}}$$

$$\textcircled{12} -\frac{9x^2}{5\sqrt{10x}}$$

$$\frac{-9x^2 \cdot \sqrt{10x}}{5\sqrt{10x} \cdot \sqrt{10x}}$$

$$\frac{-9x^2\sqrt{10x}}{5(10x)}$$

$$\frac{-9x^2\sqrt{10x}}{50x}$$

$$\frac{-9x\sqrt{10x}}{50}$$

$$\textcircled{13} \frac{-2}{7\sqrt{8x^3}}$$

$$\frac{-2}{7\sqrt{4 \cdot 2 \cdot x^2 \cdot x}}$$

$$\frac{-2}{7\sqrt{4 \cdot \sqrt{2} \cdot \sqrt{x^2} \cdot \sqrt{x}}}$$

$$\frac{-2}{7 \cdot 2 \cdot \sqrt{2} \cdot x \cdot \sqrt{x}}$$

$$\frac{-2}{14x\sqrt{2x}}$$

$$\frac{-1}{7x\sqrt{2x}}$$

$$\frac{-1 \cdot \sqrt{2x}}{7x\sqrt{2x} \cdot \sqrt{2x}}$$

$$\frac{-\sqrt{2x}}{7x(2x)}$$

$$\boxed{-\frac{\sqrt{2x}}{14x^2}}$$

$$\textcircled{14} \frac{6}{5\sqrt{18x^4}}$$

$$\frac{6}{5\sqrt{9 \cdot 2 \cdot x^4}}$$

$$\frac{6}{5\sqrt{9 \cdot \sqrt{2} \cdot \sqrt{x^4}}}$$

$$\frac{6}{5 \cdot 3 \cdot \sqrt{2} \cdot x^2}$$

$$\frac{6}{15x^2\sqrt{2}}$$

$$\frac{2}{5x^2\sqrt{2}}$$

$$\frac{2 \cdot \sqrt{2}}{5x^2\sqrt{2} \cdot \sqrt{2}}$$

$$\frac{2\sqrt{2}}{5x^2(2)}$$

$$\frac{2\sqrt{2}}{10x^2} = \boxed{\frac{\sqrt{2}}{5x^2}}$$

$$\textcircled{15} \sqrt{\frac{5}{81x^6}}$$

$$\frac{\sqrt{5}}{\sqrt{81x^6}}$$

$$\frac{\sqrt{5}}{\sqrt{81} \sqrt{x^6}}$$

$$\frac{\sqrt{5}}{9 \cdot x^3}$$

$$\boxed{\frac{\sqrt{5}}{9x^3}}$$

$$\textcircled{16} \sqrt{\frac{11}{36x^2}}$$

$$\frac{\sqrt{11}}{\sqrt{36x^2}}$$

$$\frac{\sqrt{11}}{\sqrt{36} \cdot \sqrt{x^2}}$$

$$\frac{\sqrt{11}}{6 \cdot x}$$

$$\boxed{\frac{\sqrt{11}}{6x}}$$

$$\textcircled{17} \sqrt{\frac{49}{2x}}$$

$$\frac{\sqrt{49}}{\sqrt{2x}}$$

$$\frac{7}{\sqrt{2x}}$$

$$\frac{7 \cdot \sqrt{2x}}{\sqrt{2x} \cdot \sqrt{2x}}$$

$$\boxed{\frac{7\sqrt{2x}}{2x}}$$

$$\textcircled{18} \sqrt{\frac{25}{7x}}$$

$$\frac{\sqrt{25}}{\sqrt{7x}}$$

$$\frac{5}{\sqrt{7x}}$$

$$\frac{5 \cdot \sqrt{7x}}{\sqrt{7x} \cdot \sqrt{7x}}$$

$$\boxed{\frac{5\sqrt{7x}}{7x}}$$

$$\textcircled{19} \sqrt{\frac{5}{6x}}$$

$$\frac{\sqrt{5}}{\sqrt{6x}}$$

$$\frac{\sqrt{5} \cdot \sqrt{6x}}{\sqrt{6x} \cdot \sqrt{6x}}$$

$$\frac{\sqrt{5 \cdot 6x}}{6x}$$

$$\boxed{\frac{\sqrt{30x}}{6x}}$$

$$\textcircled{20} \sqrt{\frac{2x}{7}}$$

$$\frac{\sqrt{2x}}{\sqrt{7}}$$

$$\frac{\sqrt{2x} \cdot \sqrt{7}}{\sqrt{7} \cdot \sqrt{7}}$$

$$\boxed{\frac{\sqrt{14x}}{7}}$$

$$\textcircled{21} \sqrt{\frac{50x^7}{13}}$$

$$\frac{\sqrt{50x^7}}{\sqrt{13}}$$

$$\frac{\sqrt{25 \cdot 2 \cdot x^6 \cdot x}}{\sqrt{13}}$$

$$\frac{\sqrt{25} \cdot \sqrt{2} \cdot \sqrt{x^6} \cdot \sqrt{x}}{\sqrt{13}}$$

$$\frac{5 \cdot \sqrt{2} \cdot x^3 \cdot \sqrt{x}}{\sqrt{13}}$$

$$\frac{5x^3\sqrt{2x}}{\sqrt{13}}$$

$$\frac{5x^3\sqrt{2x} \cdot \sqrt{13}}{\sqrt{13} \cdot \sqrt{13}}$$

$$\boxed{\frac{5x^3\sqrt{26x}}{13}}$$

$$\textcircled{22} \sqrt{\frac{32x^5}{7}}$$

$$\frac{\sqrt{32x^5}}{\sqrt{7}}$$

$$\frac{\sqrt{16 \cdot 2 \cdot x^4 \cdot x}}{\sqrt{7}}$$

$$\frac{\sqrt{16} \cdot \sqrt{2} \cdot \sqrt{x^4} \cdot \sqrt{x}}{\sqrt{7}}$$

$$\frac{4 \cdot \sqrt{2} \cdot x^2 \cdot \sqrt{x}}{\sqrt{7}}$$

$$\frac{4x^2\sqrt{2x}}{\sqrt{7}}$$

$$\frac{4x^2\sqrt{2x} \cdot \sqrt{7}}{\sqrt{7} \cdot \sqrt{7}}$$

$$\boxed{\frac{4x^2\sqrt{14x}}{7}}$$

$$\textcircled{23} \sqrt{\frac{10}{27x^3}}$$

$$\frac{\sqrt{10}}{\sqrt{27x^3}}$$

$$\frac{\sqrt{10}}{\sqrt{9 \cdot 3 \cdot x^2 \cdot x}}$$

$$\frac{\sqrt{10}}{\sqrt{9} \cdot \sqrt{3} \cdot \sqrt{x^2} \cdot \sqrt{x}}$$

$$\frac{\sqrt{10}}{3 \cdot \sqrt{3} \cdot x \cdot \sqrt{x}}$$

$$\frac{\sqrt{10}}{3x\sqrt{3x}}$$

$$\frac{\sqrt{10} \cdot \sqrt{3x}}{3x\sqrt{3x} \cdot \sqrt{3x}}$$

$$\frac{\sqrt{30x}}{3x(3x)}$$

$$\boxed{\frac{\sqrt{30x}}{9x^2}}$$

$$\textcircled{24} \sqrt{\frac{3}{28x^{10}}}$$

$$\frac{\sqrt{3}}{\sqrt{28x^{10}}}$$

$$\frac{\sqrt{3}}{\sqrt{4 \cdot 7 \cdot x^{10}}}$$

$$\frac{\sqrt{3}}{\sqrt{4 \cdot \sqrt{7} \cdot \sqrt{x^{10}}}}$$

$$\frac{\sqrt{3}}{2 \cdot \sqrt{7} \cdot x^5}$$

$$\frac{\sqrt{3}}{2x^5\sqrt{7}}$$

$$\frac{\sqrt{3} \cdot \sqrt{7}}{2x^5\sqrt{7} \cdot \sqrt{7}}$$

$$\frac{\sqrt{21}}{2x^5(7)}$$

$$\boxed{\frac{\sqrt{21}}{14x^5}}$$

$$\textcircled{25} \sqrt{\frac{8x^2}{27}}$$

$$\frac{\sqrt{8x^2}}{\sqrt{27}}$$

$$\frac{\sqrt{4 \cdot 2 \cdot x^2}}{\sqrt{9 \cdot 3}}$$

$$\frac{\sqrt{4} \cdot \sqrt{2} \cdot \sqrt{x^2}}{\sqrt{9} \cdot \sqrt{3}}$$

$$\frac{2 \cdot \sqrt{2} \cdot x}{3 \cdot \sqrt{3}}$$

$$\frac{2x\sqrt{2}}{3\sqrt{3}}$$

$$\frac{2x\sqrt{2} \cdot \sqrt{3}}{3\sqrt{3} \cdot \sqrt{3}}$$

$$\frac{2x\sqrt{6}}{3(3)}$$

$$\boxed{\frac{2x\sqrt{6}}{9}}$$

$$\textcircled{26} \sqrt{\frac{45}{44x^3}}$$

$$\frac{\sqrt{45}}{\sqrt{44x^3}}$$

$$\frac{\sqrt{9 \cdot 5}}{\sqrt{4 \cdot 11 \cdot x^2 \cdot x}}$$

$$\frac{\sqrt{9} \cdot \sqrt{5}}{\sqrt{4} \cdot \sqrt{11} \cdot \sqrt{x^2} \cdot \sqrt{x}}$$

$$\frac{3\sqrt{5}}{2 \cdot \sqrt{11} \cdot x \cdot \sqrt{x}}$$

$$\frac{3\sqrt{5}}{2x\sqrt{11x}}$$

$$\frac{3\sqrt{5} \cdot \sqrt{11x}}{2x\sqrt{11x} \cdot \sqrt{11x}}$$

$$\frac{3\sqrt{55x}}{2x(11x)}$$

$$\boxed{\frac{3\sqrt{55x}}{22x^2}}$$

$$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$$

$$\textcircled{27} \frac{\sqrt{3x^5}}{\sqrt{27x^3}}$$

$$\sqrt{\frac{3x^5}{27x^3}}$$

$$\sqrt{\frac{x^2}{9}}$$

$$\frac{\sqrt{x^2}}{\sqrt{9}}$$

$$\boxed{\frac{x}{3}}$$

$$\textcircled{28} \frac{\sqrt{8x}}{\sqrt{2x^5}}$$

$$\sqrt{\frac{8x}{2x^5}}$$

$$\sqrt{\frac{4}{x^4}}$$

$$\frac{\sqrt{4}}{\sqrt{x^4}}$$

$$\boxed{\frac{2}{x^2}}$$

$$\textcircled{29} \frac{4}{\sqrt{x}-7}$$

$$\frac{4(\sqrt{x}+7)}{(\sqrt{x}-7)(\sqrt{x}+7)}$$

$$\frac{4\sqrt{x}+28}{\sqrt{x^2}+7\sqrt{x}-7\sqrt{x}-49}$$

$$\boxed{\frac{4\sqrt{x}+28}{x-49}}$$

$$\textcircled{30} \frac{6}{\sqrt{3x}+2}$$

$$\frac{6(\sqrt{3x}-2)}{(\sqrt{3x}+2)(\sqrt{3x}-2)}$$

$$\frac{6\sqrt{3x}-12}{\sqrt{9x^2}-2\sqrt{3x}+2\sqrt{3x}-4}$$

$$\boxed{\frac{6\sqrt{3x}-12}{3x-4}}$$

$$\textcircled{31} \frac{\sqrt{2}-3}{\sqrt{5x}+1}$$

$$\frac{(\sqrt{2}-3)(\sqrt{5x}-1)}{(\sqrt{5x}+1)(\sqrt{5x}-1)}$$

$$\frac{\sqrt{10x}-\sqrt{2}-3\sqrt{5x}+3}{5x-\sqrt{5x}+\sqrt{5x}-1}$$

$$\boxed{\frac{\sqrt{10x}-\sqrt{2}-3\sqrt{5x}+3}{5x-1}}$$

$$\textcircled{32} \frac{\sqrt{6x}-2}{\sqrt{x}+9}$$

$$\frac{(\sqrt{6x}-2)(\sqrt{x}-9)}{(\sqrt{x}+9)(\sqrt{x}-9)}$$

$$\frac{\sqrt{6x^2}-9\sqrt{6x}-2\sqrt{x}+18}{x-9\sqrt{x}+9\sqrt{x}-81}$$

$$\frac{\sqrt{6} \cdot \sqrt{x^2}-9\sqrt{6x}-2\sqrt{x}+18}{x-81}$$

$$\boxed{\frac{x\sqrt{6}-9\sqrt{6x}-2\sqrt{x}+18}{x-81}}$$